# MOHAMMAD KAMRUL HASAN

kamrulhasan59246@gmail.com • Dhaka, Bangladesh
github.com/Kamrulhasan12345 • linkedin.com/in/kamrulhasan12345 • kamrulhasan12345.github.io

## Education

**Bangladesh University of Engineering and Technology (BUET)** **Apr 2025 – Present**
*B.Sc. in Computer Science and Engineering* *Dhaka, Bangladesh*

## Experience

**Backend Developer Intern** **Jun 2026 – Present**
*Version3.ai* *Dhaka, Bangladesh*

- Built the multi-tenant product database service in **NestJS**, **MongoDB** and **RabbitMQ**, keeping every store's data isolated so many merchants can share one platform safely.
- Built one-click store connection for **2** e-commerce platforms (WooCommerce, Shopify) with **OAuth 2.0** and encrypted credentials, so merchants onboard without handling API keys by hand.
- Evaluated image-search models and vector-database settings for visual product search, showing it can run on CPU-only servers instead of GPUs, with **2.2**× more searches per server.

**Software Engineering Intern** **Dec 2025 – Mar 2026**
*Edulytics* *Dhaka, Bangladesh*

- Built mobile screens in **React Native** and full-stack features in **Next.js** for an education platform's MVP.

## Projects

**LITTLETai: Faster, Lower-Energy LLM Inference on Android** [GitHub] Arm AI Optimization Challenge
*llama.cpp • Python • Shell • simpleperf • Perfetto* *2026*

- Made **llama.cpp** generate text **2.2–2.6**× **faster** with **2.6–3.3**× **less energy** on a budget Android phone by choosing the right CPU cores and thread counts.
- Traced the slowdown to threads spin-waiting on each other with **simpleperf**, and built a reproducible benchmark harness so every result can be regenerated.

**Ilham: Hadith Study Platform** [GitHub]
*PostgreSQL • Node.js • Express • TypeScript* *2026*

- Designed the **PostgreSQL** database and ETL for **14,941** hadiths, scoring each one's reliability in SQL.

**Sentinel: Offline Medical Triage App** [GitHub]
*React Native • TypeScript • TensorFlow Lite* *2025*

- Led a team of **3** building an Android app that triages symptoms fully offline with an on-device model.

**Bayyinah: Collaborative Desktop Quran App** [GitHub]
*Java • JavaFX • Spring Boot • WebRTC • SQLite* *2026*

- Built offline full-text search over all **6,236** verses and live group reading sessions with real-time audio.

## Publications

M. Rahaman, S. S. Rahman, M. S. M. Noor, K. S. S. Siddiquee, **M. K. Hasan**, et al., "From Wearable Data to Risk Literacy: Visualizing Physiological Heat Stress for Rickshaw Pullers in Dhaka," in *Proc. UbiComp/ISWC 2026, ISWC Notes & Briefs*, ACM, 2026. doi:10.1145/3830727.3834827

## Skills

**Languages:** C, C++, Python, TypeScript, JavaScript, Java, SQL, POSIX shell
**ML & inference:** llama.cpp, PyTorch, TensorFlow Lite, Keras, Hugging Face, sentence-transformers, FAISS, Qdrant, OpenCV
**Backend & data:** NestJS, FastAPI, Spring Boot, Express, WebSockets, WebRTC, PostgreSQL, MongoDB, RabbitMQ, SQLite
**Embedded & tools:** ESP32, STM32, Linux, Git, GitHub Actions, simpleperf, Perfetto, adb, Ghidra